

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python (Production Context, Testable, Maintainable Code), SQL, Javascript

Deep Learning: PyTorch, TensorFlow, Deep Learning Models, Training/Inference Paths, Feature Engineering, Retraining, Evaluation, Model Performance & Stability

MLOps: Docker, Kubernetes, CI/CD (Azure DevOps), AWS SageMaker, FastAPI, Real-time Systems, Automated Testing (pytest), Cloud Environments (Azure), Production Services

Data Engineering: Databricks, PySpark, Airflow, ETL Pipelines, Data & Training Pipelines, Data Processing

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated deep learning models for metrological applications, enhancing model performance and training efficiency for production readiness.
- Explored novel architectures and physics-informed hybrid loss functions to improve model accuracy and stability in complex systems.
- Built dashboards for data visualisation and model performance monitoring, crucial for operational insights and continuous improvement.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Improved vitiligo segmentation accuracy by 94% and increased inference speed by 5x using advanced AI algorithms, deployed with **Docker** and **FastAPI** for production.
- Developed a **real-time** tennis foul detection system with **TensorRT**, significantly reducing compute cost and processing time by 6x.
- Managed **MLOps pipelines** on **AWS SageMaker**, proactively addressing **YOLO** model data drift and maintaining **Kubernetes** infrastructure for stable operations.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Executed full-stack development of an AI-based service website using **Flask**, ensuring robust backend functionality and scalability.
- Developed an AI chatbot application leveraging the **OpenAI API** and relevant **LLM tools/frameworks**, integrating backend logic for reliable service.

Hugging Face

June 2022

Open-Source Contributor

Remote

- Co-authored the Zero-shot Image Classification technical documentation and implementation page on the Hugging Face Hub, demonstrating expertise in explaining complex ML concepts.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Built web scrapers and **ETL pipelines** for structured and unstructured data, establishing reliable data foundations for analytics.
- Automated data ingestion and cleaning workflows for client analytics use cases, preparing data for **ML models** and **training pipelines**.

Systems Ltd. - Visionet

Oct 2020 – Jan 2021

Data Engineering Intern

Lahore, PK

- Gained hands-on experience in **Hadoop ecosystem**, including **Hive, Sqoop, NiFi, PySpark, and Airflow**, for large-scale data processing.
- Authored a learning blog on data engineering concepts, demonstrating knowledge sharing and effective technical communication skills.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected a weighted-probability ensemble integrating DPFunc and DeepGOPlus, utilizing AlphaFold for feature extraction and taxonomic bias mitigation, improving model accuracy.
- Increased model accuracy by 4% through custom feature engineering across MF, CC, and BP namespaces, demonstrating deep understanding of feature impact.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Developed a **real-time** Bitcoin price forecasting system, employing advanced time-series analysis and deep learning models for predictive insights.
- Engineered data pipelines for continuous data ingestion and model inference, ensuring up-to-date predictions for operational use.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Designed real-time signal processing system integrating temporal dependencies for music generation.
- Implemented transformer-based models for sequence prediction in audio generation tasks.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Specialized in deep learning for computer vision and spatio-temporal modeling
- Relevant coursework: Advanced Neural Networks, Computer Vision, Autonomous Systems

National University of Computer and Emerging Sciences Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor
- Specialized in Machine Learning and Computer Vision with focus on time-series analysis

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python